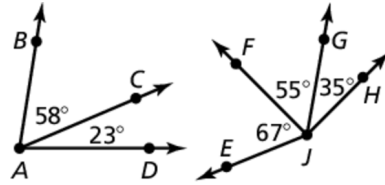


Assignment 1.9: Describing Pairs of Angles

In Exercises 1–3, use the figures.

1. Name a pair of adjacent complementary angles.
2. Name a pair of nonadjacent complementary angles.
3. Name a pair of nonadjacent supplementary angles.



In Exercises 4 and 5, find the angle measure.

4. $\angle 1$ is a complement of $\angle 2$, and $m\angle 2 = 36^\circ$. Find $m\angle 1$.
5. $\angle 3$ is a supplement of $\angle 4$, and $m\angle 4 = 75^\circ$. Find $m\angle 3$.

In Exercises 6 and 7, find the measure of each angle.

6. $\angle WXY$ and $\angle YXZ$ are supplementary angles, $m\angle WXY = (6x + 59)^\circ$, and $m\angle YXZ = (3x - 14)^\circ$.

a) Draw a picture matching the description.

b) Set up a relationship representing the description and picture you created.

c) Solve for x .

d) Find $m\angle WXY$ and $m\angle YXZ$.

7. $\angle ABC$ and $\angle CBD$ are complementary angles, $m\angle ABC = (3x + 6)^\circ$, and $m\angle CBD = (4x - 14)^\circ$.

a) Draw a picture matching the description.

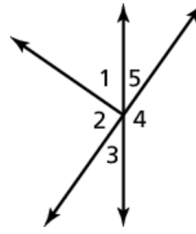
b) Set up a relationship representing the description and picture you created.

c) Solve for x .

d) Find $m\angle WXY$ and $m\angle YXZ$.

In Exercises 8–10, use the figure.

8. Identify the linear pairs that include $\angle 5$.
9. Are $\angle 3$ and $\angle 5$ vertical angles? Explain your reasoning.
10. Are $\angle 2$ and $\angle 4$ vertical angles? Explain your reasoning.



In Exercises 11–13, write and solve an algebraic equation to find the measure of each angle based on the given description.

11. Two angles form a linear pair. The larger angle's measure is 24° more than the measure of the smaller angle.
12. **Two angles are complementary.** The larger angle's measure is three times the measurement of the smaller angle.
13. **Two angles are supplementary.** The smaller angle's measure is 15 less than half the measurement of the larger angle.

14. The figure shows the design on an outdoor fence.

- a. Name a pair of adjacent supplementary angles.
- b. Name a pair of nonadjacent supplementary angles.
- c. Identify the linear pairs that include $\angle 5$.
- d. Find $m\angle 3$. Explain your reasoning.

